The Confluence of Evolutionary Computation and Multi-Agent Systems: A Survey

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***Abstract*—Both evolutionary computation (EC) and multi-agent systems (MAS) study the emergence of intelligence through the interaction and cooperation of a group of individuals. EC focuses on solving various complex optimization problems, while MAS provides a flexible model for distributed artificial intelligence. Since their group interaction mechanisms can be borrowed from each other, many studies have attempted to combine EC and MAS. With the rapid development of the Internet of Things, the confluence of EC and MAS has become more and more important, and related articles have shown a continuously growing trend during the last decades. In this survey, we first elaborate on the mutual assistance of EC and MAS from two aspects, agent-based EC and EC-assisted MAS. Agent-based EC aims to introduce characteristics of MAS into EC to improve the performance and parallelism of EC, while EC-assisted MAS aims to use EC to better solve optimization problems in MAS. Furthermore, we review studies that combine the cooperation mechanisms of EC and MAS, which greatly leverage the strengths of both sides. A description framework is built to elaborate existing studies. Promising future research directions are also discussed in conjunction with emerging technologies and real-world applications.**

***Index Terms*—Distributed artificial intelligence, distributed optimization, evolutionary computation (EC), multi-agent systems (MAS).**

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I. Introduction

In nature, simple interactions of multiple individuals can often emerge complex systematic phenomena. Interaction behaviors among individuals have attracted much attention from researchers. In the research field of artificial intelligence (AI), evolutionary computation (EC) and multi-agent systems (MAS) are two typical representatives of this phenomenon.

EC is a class of intelligent optimization algorithms that imitate natural evolution or biological behavior to solve complex optimization problems [1]–[3]. Normally, EC encodes each individual as a solution of the problem, and evolves the population through the rule of survival of the fittest in the environment. Thanks to the excellent black-box and non-convex optimization abilities, EC algorithms are widely used in combinatorial optimization, constrained optimization, multi-objective optimization, etc. [4]. MAS is a core research area of distributed artificial intelligence, providing a flexible model architecture for the distributed implementation of algorithms [5]. A multi-agent system consists of multiple autonomous agents that live and act in a given environment. Agents have the ability to sense, learn, and react in the environment. MAS has been widely used for modeling in various fields, such as sociology, industrial scenarios, and artificial intelligence [6].

The similar compositions and features of the models of EC and MAS provide the feasibility for their confluence. The confluence of EC and MAS can leverage the strengths of both sides and achieve better performance in many scenarios. On the one hand, the features of MAS, like local sensing and social interaction, can be introduced into EC. It facilitates the parallel or distributed implementation and improves the optimization effect of EC. MAS can also provide a standardized mathematical description framework for EC. Sarker and Ray published a book [7] to introduce agent-based evolutionary approaches for adaptation, learning, and optimization. Byrski *et al.* [8] introduced the basic model of evolutionary multi-agent systems and its variants. In [9], the literature is classified according to EC components represented by an agent, including solutions, heuristics, etc. On the other hand, EC does not rely on the gradient of problems and has excellent global optimization ability, which is useful to address black-box and complex problems in physical multi-agent systems. Lucidarme [10] reviewed studies that introduce evolutionary computation into multi-robot/agent systems.

During the past two decades, the number of articles that combine EC and MAS maintains a continuous growth trend, which is shown in Fig. 1(a). The growth of research interest is aided by the continuous development of technology and applications. For example, since the smart grid was proposed in the

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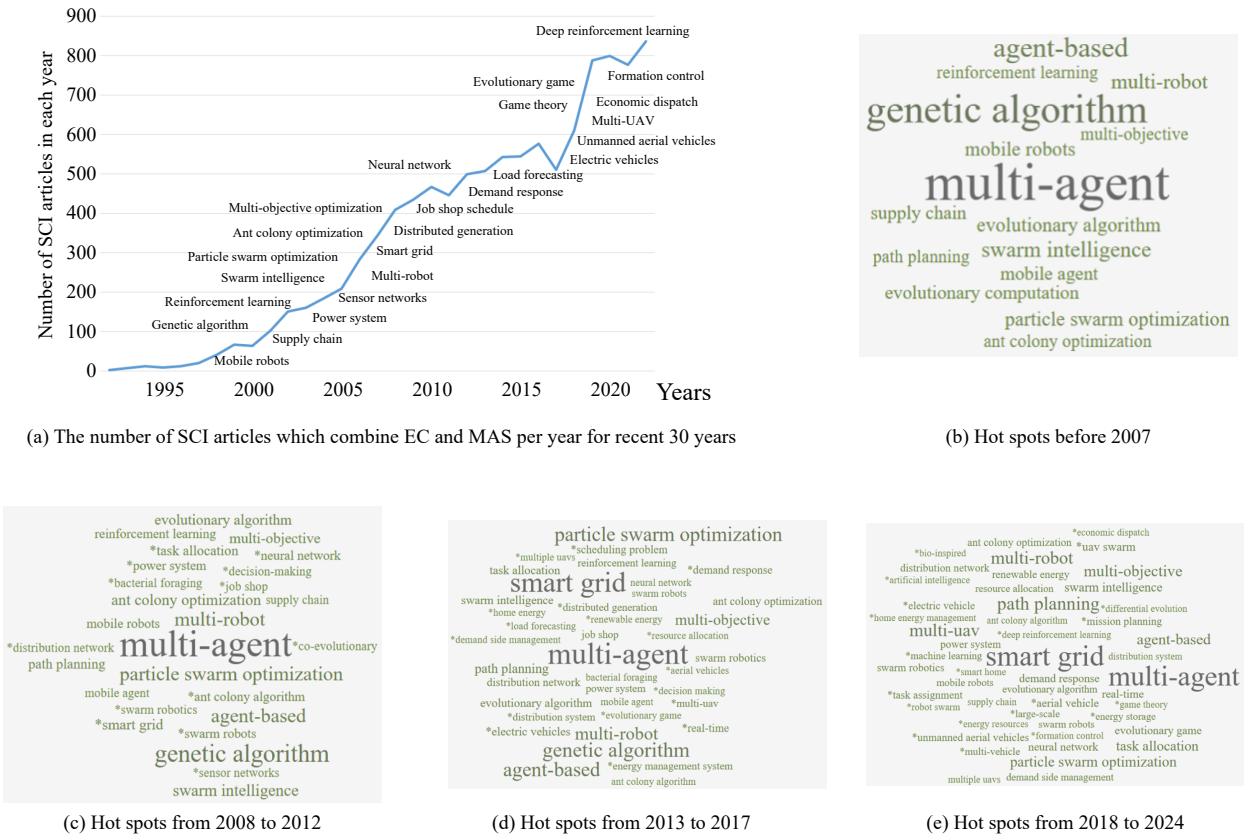

(a) The number of SCI articles which combine EC and MAS per year for recent 30 years

(b) Hot spots before 2007

(c) Hot spots from 2008 to 2012

(d) Hot spots from 2013 to 2017

(e) Hot spots from 2018 to 2024

Fig. 1. (a) The curve shows the number of related articles. The words next to the curve are the hot words at the corresponding time, which are extracted from word clouds. Among them, words above the curve are techniques and methods, while words below the curve are applications. (b-e) Hot spots in different time periods for articles combining EC and MAS. We extracted terms with more than 20 occurrences from the titles of articles in the corresponding time period. The more frequently a term appears, the larger it is in the word cloud. The symbol "*" in front of some terms indicates that the term first appears in this time period. Data comes from https://www.webofscience.com/wos/alldb/basic-search. We retrieved articles whose titles contain both MAS-related and EC-related keywords. MAS-related keywords include "multi-agent", "agent-based", "multiagent", and "multi agent". EC-related keywords include "particle swarm optimization", "evolutionary algorithm", "evolutionary computation", "swarm intelligence", "genetic algorithm", "PSO", "ant colony", "ACO", metaheuristic, "memetic algorithm", "evolution strategy", and "estimation of distribution algorithm".

2000s, EC has been widely used in distributed generation, demand response, load forecasting, energy management, etc. [11]−[13]. With the maturity of unmanned driving technology in the 2010s, EC has begun to be applied in path planning, formation control, and task allocation of multiple unmanned aerial vehicles (UAV) systems [14]−[16].

Nowadays, with the rapid development of the Internet of Things (IoT), multi-agent systems have emerged in various fields. Emerging multi-agent systems have larger scale, more complex models, and more dynamic environments. This brings both opportunities and challenges to the confluence of EC and MAS. On the one hand, with the improvement of sensing, computing, communication, and storage capabilities, terminal devices can support more complex computing methods. This gives an opportunity for EC algorithms to be applied directly on terminal devices of MAS, which can improve the real-time response and communication efficiency of the system. On the other hand, the optimization task of the emerging MAS becomes more complex, large-scale, and dynamic. It poses challenges to distributed EC algorithms that how to achieve better communication efficiency, privacy protection, robustness, and scalability in the emerging MAS. In conclusion, it is promising to further study and promote the confluence of EC and MAS.

A comprehensive and in-depth review will be helpful for further development in this field. Although there are some similar surveys in the literature, they reviewed only a subset of the confluence of EC and MAS. In detail, Carvalho and Sichman [9] only reviewed studies that use MAS to assist EC of solving optimization problems, as well as [7], [8]. Lucidarme [10] only reviewed studies that use EC to assist multi-agent systems. In conclusion, existing surveys only discuss the unidirectional assistance of EC and MAS. However, the mutual assistance of EC and MAS can play a greater role, which can not only solve complex multi-agent optimization problems, but also broaden the application scope of evolutionary computation in distributed scenarios. The above reasons drive us to review recent studies of the mutual assistance of EC and MAS, analyze their methods and progress, and propose future research directions. The contributions of this survey are summarized as follows:

1) Comparing to existing surveys [7]−[10], we have a more comprehensive classification and more recent advances for the confluence of EC and MAS. We classify existing work into three main categories based on their mutual assistance, namely, agent-based EC, EC-assisted MAS, and the intersec-

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TABLE I
COMPARISON BETWEEN THE MODELS OF EC AND MAS

	General model of evolutionary computation (EC)	General model of multi-agent systems (MAS)
Definition	A class of intelligent optimization algorithms	A kind of distributed model and architecture
	Similarities	
Composition	A population consists of multiple individuals	A system consists of multiple agents
Behaviors of entities	Evolutionary activities (crossover, mutation, etc.)	Action, sensing, reaction, social interaction, etc.
Intelligence emergence	Simple interactions of multiple individuals can solve complex optimization problems	Simple interactions among multiple agents can emerge complex systematic phenomena
	Differences	
Interaction scope	Global scope	Local scope
Goal	The population aims to solve an optimization problem	Agents act to maximize their own profits

tion of them. We also compare the characteristics, roles, and application scenarios of these three categories in detail.

2) To better review the literature, we build a description framework to introduce agent-based EC algorithms through the attributes and actions of agents. Both advantages and limitations of related works are analyzed in this survey. On the one hand, we discuss the advantages of combining EC and MAS for each category. On the other hand, we analyze the limitations of existing techniques as well as the challenges posed by emerging multi-agent systems. Further, we point out the corresponding future research directions for each challenge.

The rest of this article is organized as follows. Section II provides the background of MAS and EC. Section IV-A provides a description framework. Sections IV, V, and VI introduce three classes of existing studies respectively. Several future research direction is discussed in Section VII. Finally, we conclude this article in Section VIII.

II. BACKGROUND

In this section, we first introduce the basic concepts of EC and MAS briefly. Then, their similarity and difference are discussed.

A. Evolutionary Computation

Evolutionary computation (EC) can be mainly divided into two categories, i.e., evolutionary algorithms and swarm intelligence. Evolutionary algorithms evolve the population by the natural law of survival of the fittest, including genetic algorithms [1], genetic programming [17], differential evolution [18], evolution strategy [19]. Swarm intelligence algorithms evolve the population through simulating biological behaviors and interactions, including ant colony optimization [2], particle swarm optimization [3].

Most EC algorithms search in the solution space of the problem through the evolution of the population. A population consists of multiple individuals who interact via various evolutionary activities. For example, genetic algorithms evolve the population through mutation and crossover operations. Individuals in ant colony optimization pass information with others through a pheromone matrix. The evolution of the population cycles from generation to generation until the population converges or the maximum evaluation number is reached.

B. Multi-Agent Systems

A multi-agent system consists of multiple autonomous agents. Complex system phenomena can emerge through the learning and interaction behaviors of agents in the environment. In multi-agent systems, an agent is an entity that lives and acts autonomously in the environment. For example, an agent can be a human in a social system, a robot in an industrial system, or a virtual entity in a software system. MAS widely exists in various fields and exhibits advantages for complex system modeling and analysis.

An agent in multi-agent systems has the following characteristics:

1) Autonomous: an agent determines its own behavior and actions.

2) Goal-oriented: an agent attempts to achieve a particular goal or task.

3) Communicative: an agent can communicate with other agents.

4) Reactive: an agent can sense the local environment and respond to changes in the environment in a timely fashion.

The definition of an agent is not monolithic. There are also other characteristics of agents, such as learning ability and mobile ability. In different research fields, researchers will define the characteristics of agents specifically for different research needs.

C. Similarities and Differences

Evolutionary computation and multi-agent systems have some similar compositions and features. Table I shows the similarities and differences of their models. Their similarities are as follows. Their similarity is reflected in the composition of their models. A population in the EC algorithm contains multiple individuals, while a MAS contains multiple agents. Entities can interact with each other in the environment and are able to emerge intelligent systematic behaviors or phenomena.

Their differences are as follows. In most EC algorithms, individuals can interact with other individuals in the global scope. Differently in the MAS, agents can usually interact with neighboring agents in the local scope. Besides, agents in the MAS are autonomous to maximize their own profits. Differently in EC algorithms, individuals in the population aim to solve an optimization problem together.

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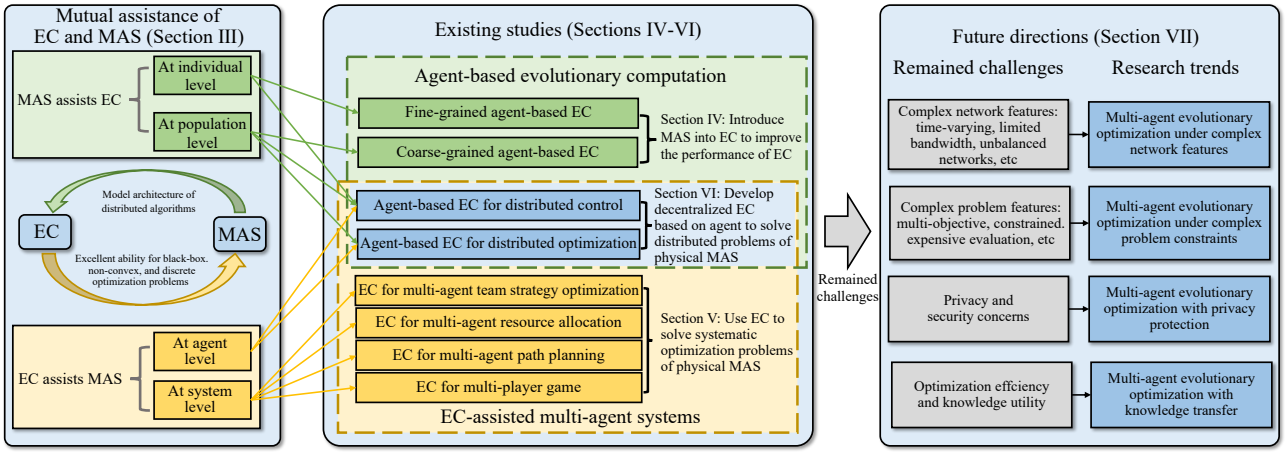
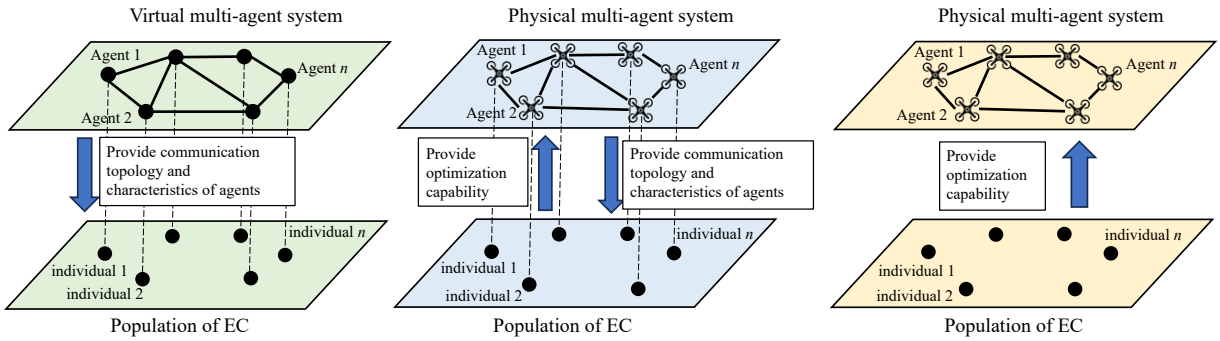


(a) Taxonomy of the confluence of EC and MAS



(b) Agent-based EC (Section IV) (c) The intersection of agent-based EC and EC-assisted MAS (Section VI) (d) EC-assisted MAS (Section V)

	Agent-based EC	The intersection of agent-based EC and EC-assisted MAS	EC-assisted MAS
Agent	Virtual entity: individuals, sub-populations	Both a physical entity and a virtual entity	Physical entity: vehicles, sensors, power stations
Topology	Virtual structure: like gird, cube, etc.	Physical communication topology	Physical communication topology: wireless sensor networks, etc.
State	Virtual state: gene, chromosome, phenomenon, etc.	Both physical states and virtual states	Physical states: position, velocity
Action	Virtual actions: self-learning (e.g., mutation), cooperation (e.g., crossover), etc.	Both physical actions and virtual actions	Physical actions: move, search, detection, etc.
Aims	Introduce MAS into EC to improve the optimization performance of EC	Develop decentralized EC based on agent to solve distributed problems of physical MAS	Use EC to solve systematic optimization problems of physical MAS

(e) Model comparison of three major classes of algorithms

Fig. 2. The taxonomy of the confluence of EC and MAS, and the model comparison of three major classes of algorithms.

III. Taxonomy

In this section, we will introduce the logic of the proposed taxonomy. According to Fig. 2(a), the taxonomy includes three parts. In the first part, we discuss two kinds of mutual assistance of EC and MAS, which are agent-based EC and EC-assisted MAS. Based on the mutual assistance relationships, we classify existing studies into three classes in the second part. Then, we list the remained challenges of existing studies and point out the future research directions.

First, in order to discuss the confluence of EC and MAS, we first introduce the mutual assistance of EC and MAS in two aspects:

1) *MAS for EC*: introduces MAS into EC algorithms at the level of individuals or populations. At the individual level, each individual is treated as an agent and placed in a cell of grid or cube structures. At the population level, an agent represents a subpopulation of EC and interacts with other agents through various communication models.

2) *EC for MAS*: adopts EC to solve complex optimization problems of MAS. EC can assist MAS at the level of agent or system. At the agent level, EC can help the agent to make autonomous optimization and control, thus enabling agents to achieve global goals through good collaboration. At the system level, EC can help MAS to solve systematic problems, such as strategy optimization, path planning, resource allocation, etc.

Existing studies may make use of one or both assistance relationships. Therefore, based on the above two kinds of mutual assistance, we classify existing studies into three classes. Figs. 2(b)−(d) illustrate the relationship between EC

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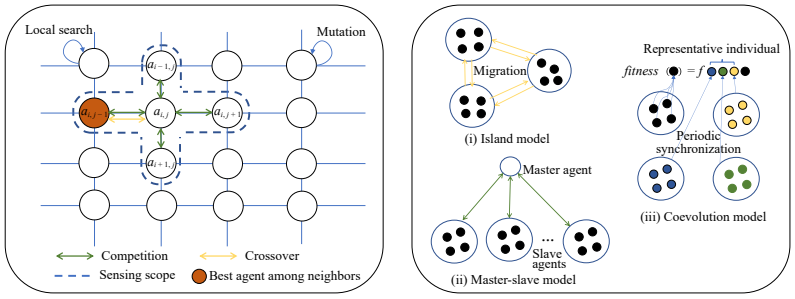

(a) Fine-grained agent-based EC (b) Coarse-grained agent-based EC

	Fine-grained agent-based EC	Coarse-grained agent-based EC
Agent	An individual	A subpopulation
Interaction frequency	Higher	Lower
Agent workload	Smaller	Larger
Number of agents	Higher	Lower

(c) Comparison

Fig. 3. Two classes of agent-based EC for centralized optimization. Fine-grained agent-based EC treats each individual in the population of evolutionary algorithms as an agent. Coarse-grained agent-based EC divides the population of EC into multiple subpopulations and treats each subpopulation as an agent.

and MAS in these three classes. Fig. 2(e) shows the difference among the three classes.

1) *Agent-based evolutionary computation:* As shown in Fig. 2(b), MAS provides communication topology and agent characteristics of agents to EC. This category aims to improve the optimization performance of EC. To be specific, each agent represents a virtual entity, corresponding to an individual or subpopulation. The topology of the system is usually defined by virtual structures, like grid, cube, etc. The states and actions of agents are virtual elements related to evolutionary algorithms. When the virtual agent represents an individual in the EC algorithm, it is termed fine-grained agent-based EC. When the virtual agent represents a sub-population in the EC algorithm, it is termed coarse-grained agent-based EC.

2) *EC-assisted MAS:* As shown in Fig. 2(d), EC provides optimization capacities for complex problems to MAS. This category of studies aims to use EC to solve systematic optimization problems of physical MAS. Existing studies mainly focus on multi-agent team strategy optimization, multi-agent resource allocation, multi-agent path planning, and multi-player game, etc. In this category, each agent represents a physical entity, such as the vehicle, sensor, and power station. The communication topology, states, and actions of agents are determined by the physical system. Take the multi-UAV system as an example, the topology depends on the communication range and the relative distance among agents.

3) *The intersection of agent-based EC and EC-assisted MAS:* As shown in Fig. 2(c), EC and MAS assist each other. On the one hand, physical MAS provides a distributed architecture and communication capability for EC algorithms. Distributed EC algorithms can be implemented and run at the level of agent. On the other hand, the developed distributed EC algorithms are used to solve distributed optimization and distributed control problems in MAS.

Although lots of researches are conducted for the confluence of EC and MAS, there still exist challenges and emerging problems. Therefore, in this taxonomy, we also list four major challenges of the confluence of EC and MAS, and the corresponding research directions.

IV. Agent-Based EC

Following the proposed taxonomy in Fig. 2(a), this section reviews studies that introduce characteristics of MAS into EC and aim to improve the performance of EC, which include fine-grained agent-based EC and coarse-grained agent-based EC. Fig. 3 shows the models and comparison of these two subclasses. In brief, each agent represents an individual in fine-grained agent-based EC, but a subpopulation in coarse-grained agent-based EC. Since the difference in granularity, agents in fine-grained agent-based EC usually have higher interaction frequency, smaller workload, and higher agent number than the coarse-grained one.

A. Description Framework

The basic idea of agent-based EC is to develop EC algorithms with the help of the autonomy and local interaction of agents. Compared with the traditional EC algorithm, agent-based EC enables the system to evolve and optimize the problem without global view and global control. In order to better show the features of agent-based EC, we use the language of MAS to define the local states and local interaction behaviors of agents. In the following, we provide a description framework to introduce agent-based EC algorithms.

The multi-agent system in agent-based EC can be described as

$$\begin{aligned} MAS &:= < \{Agent\}_{1\times n}, Topo > \\ Topo &= [\omega_{i,j}]_{n\times n} \end{aligned} \tag{1}$$

There are n agents in the system, which are connected by the topology structure $Topo$. The topology structure is described as a matrix, where $\omega_{i,j}$ is the connection between i and j.

In the simplest case, $topo$ only records whether two agents are connected, that is $topo = [\omega_{i,j}]_{n\times n} \in \{0,1\}^{n\times n}$. If the topology is weighted, it can be denoted as $topo = [\omega_{i,j}]_{n\times n} \in \mathcal{R}^{n\times n}$. If the topology is time-varying, the topology at time t is $topo^t = [\omega^t_{i,j}]_{n\times n} \in \{0,1\}^{n\times n}$.

Agents in the system can be described as

$$\begin{aligned} &Agent := < Attribute, \{Action\}_{1\times m} > \\ &\text{where} \\ &Attribute := < State, Goal, Scope, Lifespan > \\ &Action \in \{Self\text{-}Learning, Communication, \\ &\qquad Cooperation, Competition, Reaction\} \end{aligned} \tag{2}$$

where an agent consists of four attributes and m actions. With different attributes and actions of agents, the MAS can achieve different purposes. The meaning of attributes is as fol-

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lows:

1) *State*: Information of the algorithm that agents contain, such as individuals, subpopulations, or phenomena.

2) *Goal*: The purpose that drives autonomous agents to behave in the environment.

3) *Scope*: The local environment that the agent can sense and neighboring agents that the agent can communicate with.

4) *Lifespan*: Conditions under which agents live or die.

The actions of an agent fall into the following categories:

1) *Self-Learning*: Actions that an agent change and improve its *State* according to its *Goal*.

2) *Communication*: The action that the agent transfers information to neighboring agents in the *Scope*.

3) *Cooperation*: Interactive actions of agents in order to cooperate to achieve a conforming *Goal*.

4) *Competition*: Interactive actions of agents in order to obtain resources or benefits for themselves. These actions usually happen when *Goal* of agents is conflicting. The competition actions may change the *Lifespan* of agents.

5) *Reaction*: Actions that an agent responds to the request from another agent.

B. Fine-Grained Agent-Based EC

As shown in Fig. 3(a), fine-grained agent-based EC treats each individual in the population of evolutionary algorithms as an agent and introduces characteristics of MAS into traditional evolutionary algorithms. Traditional evolutionary algorithms improve the solutions of individuals through genetic operations such as crossover, mutation, and selection. Considering their limitations like premature convergence, characteristics of the agent like social behavior, autonomy, and intelligence are introduced into evolutionary algorithms. The performance of algorithms on exploration, convergence, and efficiency can be improved by introducing characteristics of MAS.

1) Model of the Multi-Agent System: The multi-agent system of this class of algorithms can be described as follows.

$$
\begin{aligned}
&MAS := < \{Agent\}_{1\times n}, Topo >\\
&Topo \in \{\text{grid, cube, ...}\}\\
&Agent := < Attribute, Action >\\
&Attribute := < State, Goal, Scope, Lifespan >\\
&Action := < Self\text{-}Learning, Cooperation, Competition >\\
&State \in \{\text{particle position, particle velocity,}\\
&\qquad \text{pheromone, gene, memetic gene, ...}\}\\
&Goal(i) := \max f(State(i)), 1 \le i \le n\\
&Scope(i) := \{(j, State(j)) | Topo_{i,j} > 0\}, 1 \le i, j \le n\\
&Lifespan := \begin{cases} 0 \text{ if agent loses competition with neighbors}\\ 1 \text{ otherwise}\end{cases}
\end{aligned} \tag{3}
$$

where f is the objective function of a maximum optimization problem, and n is the number of agents in the MAS. In the following, the attributes and actions of agents will be discussed in detail.

2) Attributes of Agents: In this class of algorithms, an agent represents an individual in the population.

State: First, an agent contains a candidate solution of the solved problem, such as a gene in genetic algorithms (GA) [20]−[22], the particle position in particle swarm optimization (PSO) [23]−[26]. In addition, an agent may contain other information in some algorithms. For example, in agent-based memetic algorithms [21], an agent contains a strategy of local search, which is also called a memetic gene. Agents will spread memetic gene with great performance to neighbors and offspring.

Sensing Scope: Compared with traditional EC where individuals can access the global environment, agents have only local sensing scope of the environment in these algorithms. Most commonly, an agent is located in a cell of the lattice structure, such as grid [20], [23] and cube [26]. In this case, the sensing scope of an agent is its surrounding four or eight neighbor agents. Each agent can sense the state of its neighboring agents and interact with them through various actions.

Goal: The basic idea of most evolutionary algorithms is that the environment selects the superior and eliminates the inferior. Therefore, the goal of each agent is to improve the solution itself and live in the environment. This goal forms a competitive relationship among agents. It is worth noting that this competition relationship is beneficial for the whole system to optimize the problem. That is because the fitness evaluation criteria of all agents are the same in centralized optimization, and the individual with the minimum fitness will be selected as the final solution.

Life-span: Normally, for agent-based genetic algorithms [20]−[22], the life-span of an agent is decided by its performance when competing with other agents. For example, if an agent loses the competition with its neighbors, it will be replaced by a new agent [20]. The new agent is generated according to the individual with the highest energy among its neighbors. Besides, the life-span of the particle is permanent in some algorithms [25].

3) Actions of Agents: An agent is intelligent and social, which may improve itself by self-learning and interaction with its neighboring agents. Therefore, various actions of agents are introduced to agent-based EC to improve algorithm performance.

Self-learning: Agents will improve the solution itself through self-learning. The most common methods for self-learning are mutation and local search. The mutation operator is used in agent-based genetic algorithms, which randomly modifies the gene of the individual with a certain probability. Local search is a process in which agents use their knowledge to improve current solutions. For example, in multi-agent genetic algorithm (MAGA) [20], an agent will sample a set of solutions around the current solution within a search radius. Then, the solution with the lowest fitness of the sample set will replace the original solution. In agent-based memetic algorithm (AMA) [21], there are multiple local search strategies for agents. Agents can select a strategy according to the historical performance of strategies adaptively.

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Cooperation: The cooperation operators in these algorithms are derived from traditional evolutionary algorithms, such as the crossover of GA and velocity update of PSO. However, these cooperation operators need to be adapted for multi-agent systems because agents can only sense and interact with their neighbors. For example, because the velocity update of PSO requires the global best position, agents obtain the global best solution by diffusion of topology network in APSO (agent-based PSO) [25]. To be specific, each agent sends the best solution it obtained so far to its neighbors per generation. Besides, in agent-based genetic algorithms, agents can conduct crossover with their neighbors only. In order to generate representative samples of potential individuals, the orthogonal crossover operator is adopted in MAGA [20]. Each agent conducts the crossover with its best neighboring agent and generates a set of offspring by an orthogonal array.

Competition: In addition to the cooperation between agents, agents may also compete with others in the MAS. For example, agents may compete with their neighbors on the fitness. If an agent is worse than all neighbors, it will be replaced by a new individual [20], [23]. The competition operator is beneficial to optimization because the new individual is obtained according to the solution of the best individual.

4) Advantages: There are many advantages to using fine-grained agent-based EC for centralized optimization. First, the lattice-like environment enables parallel implementation of algorithms, improving computation efficiency. Besides, the social behavior of agents can increase the diversity of algorithms and prevent premature convergence [20]. The multi-agent-based PSO [23] can converge to better solutions much faster than traditional PSO. Finally, the mathematical language of MAS is beneficial to the algorithm description and analysis of EC. For example, Milano and Roli [27] described metaheuristic algorithms with a multi-agent architecture. Byrski and Schaefer [28] provided a formal model for agent-based asynchronous evolutionary computation. Riechmann [29] described GA as a special form of an evolutionary game consisting of multiple agents, and analyzed the evolutionary dynamics of GA. Leboucher *et al.* [30] proposed an enhanced PSO integrated with evolutionary game theory, and analyzed the convergence speed theoretically.

C. Coarse-Grained Agent-Based EC

As shown in Fig. 3(b), coarse-grained agent-based EC divides the population of EC into multiple subpopulations and treats each subpopulation as an agent. This class of algorithms is usually developed for centralized optimization, where the evaluation criteria are the same for agents. Existing coarse-grained models of agent-based EC can be divided into master-slave model, island model, and coevolution model. It is worth noting that there are two types of master-slave models, one in which the slave node is responsible for fitness evaluation [31] and the other in which each salve node evolves a subpopulation [32]. We only discuss the latter one here because the former one only has a single population.

1) Model of the Multi-Agent System: The multi-agent system of coarse-grained agent-based EC is shown as follows:

$$
\begin{aligned}
&MAS := < \{Agent\}_{1\times n}, Topo > \\
&Topo \in \{\text{star, ring, complete graph, ...}\} \\
&Agent := < Attribute, Action > \\
&Attribute := < State, Goal, Scope > \\
&Action := \{Self\text{-}Learning, Cooperation\} \\
&State := \text{sub-population } \{individual\}_{1\times m} \\
&Goal(i) := \max_{individual \in State(i)} f(individual) \\
&Scope(i) := \{(j, State(j)) | Topo_{i,j} > 0\}
\end{aligned} \tag{4}
$$

where f is the objective function of a maximum optimization problem, and n is the number of agents in the MAS, and m is the size of the sub-population. Different from fine-grained agent-based EC, the state of an agent is a subpopulation, including multiple candidate solutions. The detailed attributes and actions are as follows.

2) Attributes of Agents: The goal of agents is to find an individual with the lowest fitness. And the life-span of agents is permanent in most cases.

State: In most models, each agent contains a subpopulation. Individuals in the subpopulation are encoded by decision variables of the solved problem, which is the same for each agent. In addition to population information, the master node in many master-slave models [32]–[34] contains intermediate evolution information collected from slave nodes.

Communication and sensing scope: The communication topology in most existing work, except for master-slave models, is a complete graph where any two agents can communicate with each other. Ishimizu and Tagawa [35] tested the performance of parallel differential evolution algorithm in three types of topology, ring network, torus network, and hypercube network. And it is shown that parallel differential evolution with these topology networks has a better performance than that with a complete graph.

3) Actions of Agents: During optimization, agents coevolve their subpopulations by self-learning and cooperation.

Self-learning: Agents evolve their own subpopulations at this phase by traditional ECs, such as Evolutionary Algorithm (EA) [36], [37], PSO [38], ant colony algorithm (ACO) [39], [40], differential evolution (DE) [33], etc. Agents evolve all decision variables of individuals in population-distributed models [36], [41], [42], while an agent only evolves a part of decision variables in dimension-distributed models [43]–[45]. In order to improve the efficiency and effectiveness of self-learning, researchers adjust parameters of evolution, including evolution strategies, size of subpopulations, and number of generations.

For evolution strategies, different mutation strategies are used for three types of subpopulations in [33], exploration population, exploitation population, and balance population. In [36], local search is used for the best individuals in the subpopulation and mutation operators are used for the worst ones. Besides, various heterogeneous memetic algorithms are used in different subpopulations in [46]. For the size of subpopulations, Wang *et al.* [34] proposed an adaptive granularity learning method to determine an appropriate subpopulation size

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TABLE II
EC-ASSISTED MAS

	Assistance from EC
Multi-agent team strategy optimization	The idea of co-evolution in EC is consistent with the improvement of team strategy, which helps to evolve efficient team strategy.
Multi-agent task allocation	EC can cope with complex constraints in multi-agent task allocation, such as constraints of power, speed, capacity, deadline, etc.
Multi-agent path planning	Multi-objective EC methods can balance the efficiency of multiple agents, and balance the individual efficiency and the systematic efficiency.
Multi-player games	EC can simulate the evolution process of decisions in the multi-player games.

during optimization. For the number of generations, Jia *et al.* [44] proposed a resource allocation mechanism to adjust the size and generation number of subpopulations adaptively according to the contribution of subpopulations.

Cooperation: Cooperation between agents is the key to improving the performance of these algorithms. In island models, agents cooperate through the migration of individuals from one island to another island, which helps agents exchange optimization progress with others. For example, in [8], the energy value of an individual will increase when it wins a competition, and individuals can migrate to other islands when the energy is higher than a threshold. Appropriate migration between agents can prevent subpopulations from falling into local optima, while intense migration will result in low diversity of subpopulations. The balance between them has triggered many studies, which investigate the migration frequency, individual selection, and individual replacement to improve the performance [41], [47], [48].

In coevolution models, agents cooperate through synchronizing optimization progress with others. Early work divides decision variables into mutually exclusive subcomponents [43], [49], so that agents usually exchange the best individual with others periodically. In recent years, researchers have begun to divide decision variables into overlapping subcomponents to handle non-separable large-scale problems [50] so that agents have to negotiate on the overlapping variables. The negotiation methods for overlapping variables are investigated in [51]−[53].

In master-slave models, slave agents cooperate indirectly through the master agent. The master agent aggregates information from slave agents and sends effective information to help further the evolution of subpopulations in slave agents. For example, the master node in [32] contains an archive to store the elitist individuals collected from slave nodes. And slave agents request the master agent for information about other subpopulations with an asynchronous and adaptive communication strategy. In [42], the master agent adopts a dynamic group learning strategy to allocate individuals to slave agents periodically, which can better balance diversity and convergence. In [54], the master agent adopts a superiority combination learning strategy, where each worse particle in the poor-performance agent randomly selects two well-performance agents with better particles for learning.

4) Advantages: Agent-based EC with coarse-grained models can increase population diversity and enhance the global exploration of algorithms. Dreżewski and Doroz [41] showed that the proposed agent-based co-evolutionary multi-objective algorithm can maintain a high level of population diversity and provide sustainable strategies for portfolio optimization. Besides, the communication frequency of coarse-grained models is lower than fine-grained models, making it more suitable for parallel implementation [37], [44]. Lorion *et al.* [48] showed that the proposed agent-based parallel particle swarm optimization framework accelerates the optimization, offers dynamic scalability at runtime, and fault tolerance. This class of algorithms is widely used for large-scale optimization [32], [42], [44].

D. Discussion

Agent-based EC models EC as a multi-agent system and solves optimization problems through the joint behavior of autonomous agents. In the above, we elaborate on fine-grained and coarse-grained agent-based EC respectively, which have their own advantages and applicable scenarios. In the fine-grained approach, the load of agents is more balanced and the agents cooperate more closely. The coarse-grained method has stronger independence of agents, lower communication frequency, and better scalability. Existing algorithms are designed to solve many types of problems, including global optimization [20], constrained optimization [21], multi-objective optimization [55], combinatorial optimization [22], and dynamic optimization [56]. They have been applied in various real-world applications, such as reactive power dispatch [26], software module clustering [57], network influence maximization [58], etc.

V. EC-ASSISTED MAS

EC-assisted MAS adopts EC to solve various problems in real-world multi-agent systems, such as smart grids, multiple UAV systems, and wireless sensor networks. According to the taxonomy in Fig. 2(a), we review EC-assisted MAS at the level of system, which includes multi-agent strategy optimization, multi-agent resource allocation, multi-agent path planning, and multi-player games. Table II introduces how EC provide assistance in different problems.

A. EC for Multi-Agent Team Strategy Optimization

Team strategy optimization in multi-robot systems is a challenging problem. This is because agents in the system will interact with each other and emerge complex collective behaviors. Traditional optimization methods directly search the parameter space of the team strategy, which is limited. First, if the strategies of agents are homogeneous, the team is unable to handle complex tasks because all agents have the same morphology and capability. Second, if the strategies of agents are heterogeneous, the search space of the problem is huge, and the algorithm is not scalable.

1) Assistance from EC: Facing the above difficulties, coop-

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erative co-evolutionary algorithms (CCEAs) have the potential to address multi-agent strategy optimization. Based on the idea of divide-and-conquer, CCEAs can evolve strategies of heterogeneous agents effectively by multiple cooperative sub-populations. Each sub-population searches on the parameter space of the agent strategy, which greatly decreases the search complexity and increases the scalability of the algorithm. For example, Yang *et al.* [59] used CCEAs to help physical robots with distinct capabilities synthesize their strategies and show cooperative behaviors. Collective neuro-evolution [60] used genotype and behavior specialization difference metrics to guide the emergent specialization of robots.

In order to improve the performance of team strategy optimization, many researchers analyzed and improved CCEAs in different aspects. Potter *et al.* [61] examined the tradeoff between heterogeneity and homogeneity, and found that heterogeneous agents are necessary when the number of skills needed increases. Blumenthal and Parker [62] studied how to select suitable partners for evaluation when multiple sub-populations coevolve. Waibel *et al.* [63] analyzed the genetic team composition and level of selection of CCEAs for team strategy optimization. They show that the optimal setting depends on the needed amount of collaboration.

Considering that CCEAs may converge prematurely to stable states instead of optimal solutions, a novelty-driven coevolution algorithm is proposed in [64]. It adopted novelty search that drives evolution toward novel and diverse behavior rather than exclusively pursuing a static objective. In addition, a symbiotic learning method is proposed in [65], which defines four symbiotic relationships between agents, i.e., mutualism, harm, predation, and altruism. Besides, some researchers focus on partial heterogeneity. Lichocki *et al.* [66] evolved team composition of MAS by agent swapping. It applies free agent swapping to explore first and restrict agent swapping to exploit. In [67], the agent strategies and team composition are evolved at the same time through the proposed merge and split operations among sub-populations in CCEAs.

In addition to CCEAs, the combination of EC and deep reinforcement learning (DRL) also attracts great attention for team strategy optimization [68]−[72]. Khadka *et al.* [68] proposed a sample-efficient multi-agent coordination method for multi-agent reinforcement learning, which takes the advantages of both EC and DRL. Considering the insensitivity of EC to sparse and delayed rewards, a modified genetic algorithm is used to learn the team reward. DRL is adopted to learn agent-specific rewards due to its sample efficiency. Long *et al.* [69] considered that the complexity of the environment grows exponentially with the number of agents, and proposed a curriculum learning paradigm that scales up multi-agent reinforcement learning by progressively increasing the number of agents. An evolutionary approach is designed to fix the objective misalignment throughout the curriculum. Wang *et al.* [73] use genetic programming (GP) to assist the action advising of multi-agent reinforcement learning. GP is used to find an expression tree as a trigger condition, which decides when an agent provides advice to another agent.

2) Advantages: The advantages of using EC for multi-agent team strategy optimization are as follows. First, the mapping between strategy parameters and rewards is usually black-box in many scenarios, which is difficult to optimize by gradient-based methods. As a gradient-free method, EC is beneficial to search the strategy to maximize the rewards [68]. Second, there are various methods to control population diversity of EC, which is conducive to exploring more diverse strategies [66], [67]. Third, the idea of co-evolution in EC is compatible with the improvement of team strategy, which helps to evolve efficient team strategy [59], [62].

B. EC for Multi-Agent Task Allocation

Task allocation, namely which task is assigned to which agent, remains a challenge when several tasks are conducted simultaneously in MAS. Considering task execution time, task finish rates, execution, or communication cost for agents, etc., task assignment has been proven to be a non-deterministic polynomial (NP)-hard combinatorial optimization problem [74], [75].

1) Assistance from EC: From the perspective of algorithm implementation, existing algorithms for multi-agent task allocation can be broadly divided into centralized [76], [77] and distributed [16] approaches. The centralized approaches usually keep a central agent that negotiates with all other agents, obtains all information in advance, and directly generates the solution for task assignment. The distributed approach requires multiple agents to participate in the computation and optimization for task allocation. Distributed approaches have the advantage of robustness and scalability for large-scale task allocation. However, it is much harder for distributed approaches to get consensus on task allocation among multiple agents.

From the perspective of application scenarios, existing algorithms are widely applied in task allocation for physical entities and computational tasks. In the first class, agents are usually physical entities, such as robots, UAVs, and mobile phones [78], [79]. For example, Zhou *et al.* [16] combined a distributed many-objective evolutionary algorithm with a greedy algorithm to address multi-agent task allocation. The proposed algorithm is a distributed approach with master-slave architecture, where the master agent executes population initialization, crossover and mutation operator in in the evolutionary algorithm [80] and the slave agents are responsible for the evaluation of solutions. Tao and Song [79] addressed the mobile crowdsensing problem where users are treated as workers to carry out sensing tasks, and provided a cooperative ACO to approximate the optimal solution in large-scale cases. In the second class, agents conduct computational tasks as computation resources, such as cloud service [81]–[83], cloud robotic network [84], virtual machine [85], etc. Considering the latency of the network infrastructure, dynamic topology, and node heterogeneity, Wang *et al.* [82] proposed a modified ant colony algorithm named collection path ant colony optimization (CPACO) to handle large-scale multi-agent task allocation. Guo *et al.* [84] focused on the assignment of computational tasks in a cloud robotic network (CRN) which consists of multiple mobile robots and a cloud computing center. They proposed an energy-sensitive evolutionary algorithm (ESGA) as a strategy for one robot to

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offload tasks to the cloud center and other robots, which improves computing ability and execution efficiency.

2) Advantages: Multi-agent task allocation problems often need to take into account multiple objectives at the same time, such as task completion rate, total completion time, completion cost [16]. EC has a rich research foundation in multi-objective optimization, which helps to find solutions that take into account multiple objectives [86], [87]. Besides, agents are often heterogeneous in terms of power, speed, capacity, type, deadline, etc., which makes the task allocation have multiple complex constraints at the same time. EC has a lot of research on constrained optimization, which is helpful to solve constrained task allocation.

C. EC for Multi-Agent Path Planning

Multi-agent path planning problems aim to plan the path for multiple agents synchronously and minimize the system cost [88]. Compared with normal path planning problems, the major challenges of multi-agent path planning lie in the following three points. First, the trajectories of agents affect each other. Second, each agent only has a local observation of the environment. Third, the environment is dynamic for an agent due to the synchronous actions of multiple agents.

1) Assistance from EC: Existing studies for MAS path planning can be divided into two categories, offline planning and online planning. Algorithms in the first category optimize the path of each agent offline, and then the agent executes according to the dispatched path. For example, Das *et al.* [89] consider optimization problems of the heterogeneous vehicles routing with time windows. A collaborative pareto ACO algorithm is proposed to optimize the path and order of agents offline algorithm is used to optimize the path and order of agents offline. Jiang *et al.* [90] studied the vehicle routing in the scene of maritime search and rescue. The group control agent first models the search area according to electronic charts and weather information, and then uses an improved ACO to find the shortest traveling path for each rescue agent. Besides, the routing problem of network packets also belongs to MAS path planning problems. Seyyedabbasi and Kiani [91] use the ACO algorithm to optimize the routing table of routers in the sensor network, which obtains appropriate and effective paths for data transmission and reduces the overall energy consumption of the network.

Studies about online planning consider the dynamic environment that changes during the execution of agents [92], [93]. In this case, agents need to avoid collisions with obstacles or other agents. Multi-agent pathfinding (MAPF) is a typical problem that aims to find the shortest possible path for each agent without collisions with other agents or obstacles [94]. For example, in [95], the agent will recalculate its path to avoid collisions when an obstacle is found. Yiu and Mahapatra [94] use A* algorithm to conduct real-time path planning according to environmental information. And GA is used to optimize the heuristic function of A* algorithm to improve the search ability and accuracy.

In addition to being used for path planning, EC can also assist other algorithms. First, EC can be used to optimize algorithm parameters. For example, Liu *et al.* [96] developed a genetic algorithm based K-means algorithm, which uses a genetic algorithm to optimize cluster center for the K-means algorithm. Second, EC can be used for path planning at a coarse granularity. Yu *et al.* [97] use DE and ACO to plan the voyage at a coarse granularity and then use A* algorithm to avoid collisions at a fine granularity.

2) Advantages: The advantages of using EC for multi-agent path planning are as follows. First, facing online path planning tasks, the real-time and interpretability of vehicle decisions are very important. Genetic programming in EC make the decision by a rule tree, which is useful for this problem [98], [99]. On the one hand, it can express decision function explicitly, and on the other hand, the result can be calculated in linear time. Second, path planning needs to balance the efficiency of multiple agents, and balance the individual efficiency and the systematic efficiency. Multi-objective evolutionary algorithm is beneficial to solve this kind of multi-objective problem. For example, [100] shows the effectiveness of using the genetic algorithm to address the multi-objective multi-agent path planning problem.

D. EC for Multi-Player Games

A multi-player game involves a scenario where multiple autonomous agents interact by taking actions to maximize their own profits within an environment. The action of each agent influences the profits and payoffs not only for itself but also for other agents involved in the game. Compared to two-player games, the multi-player game tends to be more high-dimensional, dynamic, and uncertain.

1) Assistance from EC: EC is usually used to find equilibrium points for multi-player games [101], [102] or solve complex optimization problems containing multi-player games [103]−[106].

For the first class, Wang *et al.* [102] adopted PSO for strategy updating in the prisoner dilemma and the snowdrift game, and showed that PSO can promote cooperation among agents even under unfavorable conditions in both games. In [101], an evolutionary algorithm, colonial competitive algorithm, is used to find Nash equilibrium points of nonlinear non-cooperative games.

For the second class, Neshat and Amin-Naseri [104] proposed an agent-based hybrid framework of game theory and particle swarm optimization for market-driven generation expansion planning. This work considered the uncertainty and dynamics of the market prices affected by multiple supply agents and demand agents. PSO is adopted to optimize the decision for each individual based on the current market price during the game. Similarly in [105], PSO is used for decision-making of investment in a multi-agent game. In [106], the energy market is modeled by a Stackelberg game consisting of one leader and multiple followers. Considering the game is nonlinear, nonconvex, and constrained, a unified PSO is used to optimize the decisions for customers.

2) Advantages: In the multi-player game, individuals choose actions based on their own interests, and adjust their strategies according to the game results. The evolution mechanism of EC is similar to the process of game, which is conducive to more closely imitating the evolution process of the system.

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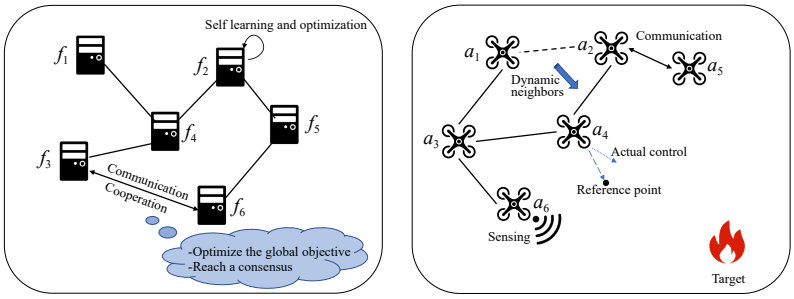

(a) Agent-based EC for distributed optimization (b) Agent-based EC for distributed control

	AEC for distributed optimization	AEC for distributed control
Goal	Find the optimal solution satisfying certain constraints	Perform formation, tracking or other tasks
Environment interaction	Agents evaluate solution quality by the environment information	Agents act in the environment and get feedback
Environment dynamics	Relatively static	Relatively dynamic

(c) Comparison

Fig. 4. Two classes of agent-based EC for multi-agent systems. Distributed optimization aims to optimize a global objective function which is usually defined by the sum of multiple local objective functions. Distributed control aims to achieve a common goal through the cooperation of mobile vehicles or robots.

For example, Zhang *et al.* [107] proposed an individual evolutionary game model guided by global evolutionary optimization for vehicle energy station distribution. This work aimed at finding a policy for the government to motivate selfish investors to adopt strategies in accordance with the overall interests of all customers.

E. Discussion

EC-assisted MAS utilizes the optimization ability of EC to solve the systematic optimization problems of practical MAS. EC has three main advantages to solving practical problems in MAS. First, EC does not rely on gradient information, so it can handle black-box problems when the mathematical model of the problem is unknown, such as team strategy optimization. Second, EC has rich research foundations on multi-objective optimization, constrained optimization, real-time decision, diversity search, etc. These can handle complex problem features in real-world MAS problems. Third, it is convenient to use EC to model various discrete problems, which plays an important role in task allocation and path planning.

VI. The Intersection of Agent-Based EC and EC-Assisted MAS

With the development of communication technology, networked systems have emerged in various fields. In many emerging scenarios, edge devices or terminal hardware are gradually equipped with better computing and communication capabilities. This provides a basis for the development of distributed EC algorithms. Meanwhile, data related to the optimization problems in emerging MAS scenarios is often sensed and stored in terminal devices. It is not practical to aggregate global information from all agents for centralized optimization due to data privacy, communication cost, robustness, and computation efficiency. Therefore, the distributed optimization and control of MAS is of great challenge and significance.

Facing the above challenges, many studies aim to combine the cooperation mechanism of EC and MAS. To be specific, they develop distributed evolutionary algorithms via the characteristics of MAS and try to solve the distributed optimization and control problems of MAS. This class of studies is the intersection of agent-based EC and EC-assisted MAS. On the one hand, physical MAS provides a distributed architecture and communication capability for EC algorithms. On the other hand, the developed distributed EC algorithms are used to solve distributed optimization and distributed control problems in MAS.

A. Agent-Based EC for Distributed Optimization

In recent years, researchers have developed many distributed evolutionary algorithms to solve distributed optimization problems. Existing work mainly addresses three types of distributed optimization problems, i.e., objective distributed problems [108]–[110], problem-information distributed problems [111]–[113], data distributed problems [114]–[116]. In objective distributed problems, each agent has a local objective function and the global objective function is a sum of all local objective functions. In problem-information distributed problems, the system needs to complete a task, but the nodes do not know each other's capabilities, status, or communication range. In data-distributed problems, the data for optimization is stored in different nodes.

1) Model of the Multi-Agent System: The multi-agent system of this class of algorithms is defined in the following.

$$
\begin{aligned}
&MAS := < \{Agent\}_{1\times n}, Topo > \\
&Topo \in \{\text{connected graph}\} \\
&Agent := < Attribute, Action > \\
&Attribute := < State, Goal, Scope, Lifespan > \\
&Action := \{Self\text{-}Learning, Cooperation, Communication\} \\
&State := \text{sub-population } \{individual\}_{1\times m} \\
&Goal := \max \sum_{i=1}^{n} f_i(State(i)) \\
&Scope(i) := \{(j, State(j)) | Topo_{i,j} > 0\}
\end{aligned} \tag{5}
$$

Compared with fine-grained or coarse-grained agent-based EC, the goal of this class is the sum of multiple local objective functions f_i $(i = 1, 2, ..., n)$. Because each agent can only access its own local objective function or local data, this goal requires the cooperation of all the agents.

2) Attributes of Agents: As shown in Fig. 4(a), each agent normally represents a node in the networked system, such as a station in electric power systems [117], [118] or a sensor in wireless sensor networks [111], [119]. Algorithms run on these entity nodes and distributed optimization is achieved by local computation and communication.

State: Normally, an agent contains a subpopulation [108],

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[120], in which each individual is encoded by the decision variables of the problem. In addition to subpopulations, agents may contain other information, such as pheromone of ACO-based algorithms [111], [121], [122] and surrogate models [114], [115].

Goal: The ultimate goal of agents is to cooperate to optimize the global objective. This goal is challenging because each agent can only access a part of the problem information. For example, local objective functions of agents are usually in conflict in objective-distributed problems.

Life-Span: As the above introduction, each agent corresponds to a node in networked systems so that their life-span is permanent during optimization.

Communication and Sensing Scope: Limited to the communication distance, agents often communicate with neighboring agents in the system. In the most common case, neighbors of an agent depend on the geographic distance between agents [113], [117], [123]. Besides, some algorithms design the communication topology specifically, such as the tree structure [124] and link structure [125].

3) Actions of Agents: Two key challenges of distributed optimization are how to optimize the global objective and how to reach a consensus among agents. In the optimization process, there are three main actions of agents, that is self-learning, communication, and cooperation.

Self-learning: At the phase of self-learning, agents evolve their subpopulations by classical ECs, such as PSO [120], [123], [124], [126], EA [112], [113], and ES [110]. In order to consider the global objective better, some methods to guide self-learning are proposed. For example, the velocity update strategy of PSO in [126] is adapted by adding an extra item, which enables particles to learn from the best solutions from other agents. In [117], [120], an auxiliary variable is used to help agents estimate global fitness at the phase of self-learning.

Communication: At the phase of communication, agents exchange various kinds of information with neighbors, such as individuals of the subpopulation [108], [111], [127], auxiliary variables [117], [120], fitness of solutions [108], [124], [125]. The communication may be simultaneous or sequential. In the case of simultaneous communication [108], [120], agents exchange new information found in the phase of self-learning. In the case of sequential communication [124], [125], an agent receives the information from the downstream agents, integrates it into its own information and sends it to the upstream agent.

Cooperation: Cooperation among agents is important to address the two above-mentioned challenges, how to optimize the global objective and how to reach a consensus.

Aiming at optimizing the global objective, there are three ways in the existing work, computing the global fitness of individuals, estimating the global fitness of individuals, and incorporating solutions of different agents. First, computing global fitness requires the integration of local fitness evaluated by all agents. The integration is achieved by repeated weighted averaging in [108] or accumulation from downstream agents to upstream agents in [124], [125]. Second, estimating global fitness requires auxiliary variables [120] or surrogate models [114] trained by historical experience. Third, many algorithms incorporate solutions of different agents [112], [123], [126]. For example, in [123], the local best individual of an agent will be replaced if the best individual of neighbors is better than itself.

Aiming at reaching a consensus among agents, weighted averaging based on the consensus theory is most commonly used [108], [110], [120]. For example, in [108], agents weighted the average of subpopulation of themselves and neighboring agents. After a certain number of averaging, the system will reach a consensus according to the consensus theory. A negotiation approach for distributed task allocation is proposed in [113], where agents decide whether to accept a task by comparing the reduced cost with the increased cost.

4) Advantages: The advantages of agent-based EC for distributed optimization lie in the following three points. First, agent-based EC can solve distributed black-box problems, while most existing distributed optimization algorithms rely on the gradient of the objective function. Chen *et al.* [109] have verified the effectiveness of the proposed multi-agent coevolutionary algorithm on an electric dispatching problem involving game, whose objective function is gradient-infeasible. Second, agent-based EC has a powerful search ability for non-convex problems. Wang and He [123] demonstrated that the proposed algorithm outperforms distributed gradient search for target localization problems in wireless sensor networks. Third, agent-based EC is suitable to address discrete problems, which widely exist in real-world applications [111], [113], [128]. Qiu *et al.* [128] applied the proposed coevolutionary algorithm in a network-based combinatorial optimization problem, i.e., multidepot vehicle routing problem with time windows.

B. Agent-based EC for Distributed Control

Agent-based EC for distributed control has attracted lots of attention from researchers. Distributed control is very common in multiple UAV or multi-robot systems, such as the source location task and target tracking task. It is challenging that agents not only need to optimize objective functions but also ensure movement safety and stability in an actual environment. Therefore, these algorithms usually combine swarm intelligence and control theory to achieve effective distributed control [129], [130].

1) Model of the Multi-Agent System: Equation (6) describes the multi-agent system of this class of algorithms.

$$\begin{aligned}
&MAS := < \{Agent\}_{1\times n}, Topo >\\
&Topo \in \{\text{time-varying graph}\}\\
&Agent := < Attribute, Action >\\
&Attribute := < State, Goal, Scope, Lifespan >\\
&Action := \{\text{Sensing, Communication, Reference}\\
&\qquad\qquad \text{position selection, Control signal output}\}\\
&State \in \{\text{position, velocity, sub-population}\}\\
&Goal \in \{\text{location task, formation task, ...}\}\\
&Scope(i) := \{(j, State(j)) | distance(State(i), State(j))\\
&\qquad\qquad < \text{maximum communication distance}\}
\end{aligned} \tag{6}$$

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In distributed control problems, agents are usually mobile robots or vehicles. In this case, the sensing of agents is of great importance, and the topology network is time-varying according to the positions of agents.

2) Attributes of Agents: As shown in Fig. 4(b), an agent represents a mobile entity, such as a vehicle or a robot.

State: Firstly, an agent contains its actual position and velocity in the environment. Then, for intelligent planning, it also contains the parameters of swarm intelligence algorithms. It may contain an individual in fine-grained models [129], [131]–[134] or a subpopulation in coarse-grained models [112], [113], [135]–[137].

Goal: In these algorithms, the ultimate goal of agents is the same in cooperative tasks. For example, they may aim to find a position with the highest odor concentration or a position closest to the target. Besides, agents may consider additional goals during the running process, such as the safety, stability, and group formation of movement. The fitness or cost function differs depending on the tasks. For target localization, the fitness of a solution is evaluated by odor concentration [132], [133] or target distance [129], [136]. For group formation, the fitness of a solution is evaluated by the deviation from the desired formation [138], [139].

Life-span: As the above introduction, agents often represent real entities like vehicles and robots, so that their life-span is permanent during optimization.

Communication and sensing scope: In the actual environment, the communication and sensing scope of agents depends on the communication radius of the represented entity. The neighboring agents are changed dynamically because agents are mobile in the environment.

3) Actions of Agents: During the task execution, there are three main actions of agents. First, agents will sense the environment and communicate with neighbors through various sensors. Then, agents select a reference point, or a desired position, for the next step according to collected information. Finally, agents output the control signal to entities according to the reference point.

Sensing and communication: Agents will sense the state of environment by various types of sensors, such as radar sensors for distance detection [134], [135], infrared sensor for temperature detection [136], odor sensor for concentration detection [131]–[133]. These sensors are applied to detect targets, neighboring teammates, and obstacles. Agents usually cooperate with each other by communicating detection results and their local information, which helps agents to conduct movement decisions autonomously. For example, at target localization tasks, agents will transfer the best position they found so far to their neighboring agents [132], [134].

Reference position selection: The cooperative control of autonomous agents in these algorithms is mainly based on swarm intelligence and cooperative control theories. PSO is the most commonly used swarm intelligence method for distributed target localization. That is because PSO is inspired by bird foraging and is suitable for swarm cooperative localization in the environment. Classical PSO evolves particles according to personal best position and global best position. In the actual environment, the personal best position is recorded in local entities, and the global best position is obtained through information sharing among agents.

Except for algorithms that use classical PSO [132], [138] to select reference points, many researchers have modified PSO to improve the actual control effect. For example, Jatmiko *et al.* [131] proposed a charged PSO inspired by Coulomb's law for odor source localization, where any two agents will generate a repulsive force if their distance is less than a threshold. multi-agent coordination optimization (MCO) [129], [134] reconstructs velocity update rule of PSO with dynamic consensus protocols and a gradient-based adaptation. In [133], PSO is used to compute the oscillation center through formula translation. In addition to PSO, other swarm intelligence methods, like DE [136] and pigeon hierarchical strategy [140], are also adopted for distributed control.

Control signal output: In the ideal situation, the agents can reach the reference point directly. However, the actual movement of entities needs to take into account obstacle avoidance, maximum torque, etc. Therefore, many approaches are proposed to confirm the actual movement of agents. For example, a control law is adopted in [132] for control signal output, which considers not only the reference point of the agent itself but also that of neighboring agents. The control law can guarantee that the actual position of robots will converge to the expected position. In [135], agents generate a smooth path from the current position to the reference point by a fifth-order polynomial. In [133], agents plan the trajectory in three possible ways, surging, casting, and exploration.

4) Advantages: Since EC is inspired by the biological behavior in nature, agent-based EC can provide better intelligence and collaboration for distributed control tasks. For example, Luo and Duan [140] designed a hierarchical strategy for UAV flocking control inspired by pigeon formation behavior. The team stability and efficiency of pigeon-inspired flocking control method are verified through simulations. In addition, Jatmiko *et al.* [131] also demonstrate that the proposed PSO-based mobile robot strategy can detect and respond to the dynamic environmental changes for odor source localization tasks.

C. Discussion

Combining the cooperation mechanism of EC and MAS, this type of method can take both advantages better. To summarize the above two subclasses, agent-based EC for distributed optimization usually aims to find the optimal solution satisfying certain constraints, while distributed control aims to perform formation, tracking, or other tasks. For distributed optimization, the environment is relatively static. Agents evaluate solution quality by the environmental information. For distributed control, the environment is relatively dynamic. Agents act in the environment and get feedback.

VII. Current Summary and Future Directions

According to the literature review in the above sections, the confluence of EC and MAS has achieved lots of research results. On the one hand, MAS can provide model architectures and platforms to assist the distributed implementation of EC algorithms. On the other hand, EC can assist MAS by

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solving complex optimization problems. Currently, related methods have been studied and applied in various real-world scenarios. According to Fig. 1(a), the application scenarios of these methods are closely related to the development of real multi-agent systems. For example, supply chain applications were widely studied before 2007, while smart grids and multi-UAV systems have attracted much attention in recent years. In the future, with the development of the IoT, the optimization requirements of realistic multi-agent systems will continue to grow.

Faced with the complexity of real-world systems and the growth of optimization requirements, existing methods face the following challenges. First, facing time-varying, limited bandwidth, or unbalanced networks, it is necessary to research multi-agent evolutionary optimization under complex network features in the future. Second, multi-agent evolutionary optimization under complex problem features is worth further research due to multi-objective, constrained, expensive evaluation problems. Third, privacy security is concerned in real-world scenarios, it is promising to propose multi-agent evolutionary optimization with privacy protection. Finally, considering the optimization efficiency and utility of knowledge, multi-agent evolutionary optimization with knowledge transfer is also one of important trends. These challenges make this field still have great research value and room for progress. In the following, we will discuss these challenges and point out the corresponding future research directions.

A. Multi-Agent Evolutionary Optimization Under Complex Network Features

In the real-world multi-agent systems, the network features are usually complex, such as time-varying networks, limited bandwidth networks, and unbalanced networks. Although EC has begun to be used to solve black-box and non-convex multi-agent optimization problems, existing studies mainly address static and balanced communication networks without bandwidth limitation [108], [127]. First, agents are mobile in the multi-vehicle or multi-robot systems, which makes the communication network dynamic based on the distance among agents [141]. Second, in the wireless sensor networks, the communication bandwidth is limited by the hardware, which poses challenges to the lightness of evolutionary optimization algorithms [142]. Third, in the traffic networks, the flow and requirement of agents are usually unbalanced, which also poses challenges to the resource dispatch of evolutionary optimization algorithms [143].

B. Multi-Agent Evolutionary Optimization Under Complex Problem Features

According to research hotspots in Fig. 1, MAS widely exists in real-world applications, including road networks, smart grids, social networks, industrial chains, etc. Although EC has shown advantages on black-box optimization, existing studies seldom consider multi-objective, constrained, evaluation-expensive condition for distributed optimization problems. First, considering a scenario where multiple companies collaborate, each company owns only one or part of objective functions. It is worth researching how to develop multi-agent evolutionary optimization method to find the Pareto solutions [144]. Second, in a hyperparameter optimization problem for distributed machine learning, the evaluation of hyperparameters is computational expensive. How to develop multi-agent evolutionary optimization method to find an ideal hyperparameter with low computation cost is worth studying [145].

C. Multi-Agent Evolutionary Optimization With Privacy Protection

Nowadays, the data privacy of governments, enterprises, and users is critical. In emerging computing scenarios, such as mobile crowdsensing, edge computing, and cloud computing, multiple computing agents are required to collaborate on optimization or learning tasks under the promise of data security. However, only a few studies consider privacy protection of multi-agent evolutionary optimization methods [146], [147]. The privacy topic is not fully researched yet. It is of great significance to develop privacy-preserving multi-agent evolutionary optimization for various real-world applications [148].

D. Multi-Agent Evolutionary Optimization With Knowledge Transfer

Optimization efficiency and knowledge utility are important of multi-agent optimization, which is also the limitation of existing methods. At present, scholars have begun to study the knowledge transfer in multi-agent evolutionary optimization. For example, Hou *et al.* [149] proposed an evolutionary transfer reinforcement learning (eTL) framework for multi-agent systems, enabling agents with higher rewards to transfer experience to other agents. The research focuses on how to identify suitable source tasks from multiple agents and how to transfer structured knowledge effectively [150]−[152]. Recently, multi-agent frameworks for large language model (LLM) have received extensive attentions, which aim to develop an effective cooperation mode for multiple LLM agents [153]−[155]. The interaction mechanism in EC has great potential to be applied to multi-agent LLM frameworks, because EC has similarities with the biological interaction behavior in nature.

VIII. Conclusion

In this survey, we provide a detailed review of the mutual assistance of EC and MAS. First, the characteristics of MAS can be introduced into EC to improve the performance and parallelism of EC. Second, EC can be adopted to solve MAS optimization problems. Third, the cooperation mechanism of EC and MAS can be combined to better solve distributed optimization and control problems. Nowadays, with the development of IoT and edge computing, it is more and more important to consider emerging optimization problems of MAS. As the future research directions we outlined, the confluence of EC and MAS is of great research value.

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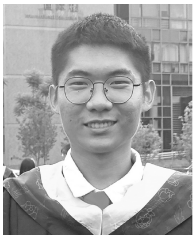

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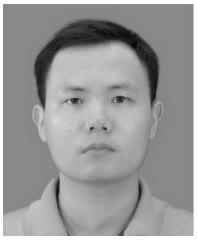

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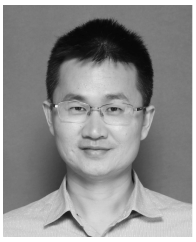

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